

Self-Learning: NoSQL Data Models -- DBMS

Definition: NoSQL (Not Only SQL) refers to non-relational database management systems designed for horizontal scalability, high-velocity big data, flexible schema-less data models, and high availability.

1. Detailed Technical Explanation

NoSQL databases trade traditional ACID strict transactional guarantees for high performance and horizontal scaling across distributed clusters (governed by the CAP Theorem).

CAP Theorem:

A distributed database system can guarantee at most two out of three properties simultaneously:

1. Co

2. The Four Major NoSQL Data Models

NoSQL Category	Data Model	Key Features	Popular Industry Tools	Use Cases
Key-Value Store	Hash Table (Key -> Value Blob)	Fast lookup by key, ultra-high p	Redis, Amazon DynamoDB, Ri	Caching, session management, user sh
Document Store	JSON / BSON / XML Document	Schema-free nested documents	MongoDB, CouchDB	Content management, e-commerce proo
Column-Family Store	Sparse tables indexed by Rows	Optimized for heavy write throu	Apache Cassandra, HBase	Time-series data, IoT analytics, finan
Graph Database	Nodes, Edges (Properties)	Graph traversal queries for inter	Neo4j, Amazon Neptune	Social networks, fraud detection, reco

3. SQL vs NoSQL Architecture Comparison

Feature	Relational SQL Databases	NoSQL Databases
Data Schema	Fixed, predefined rigid schema.	Dynamic, schema-less / flexible.
Scaling	Vertical scaling (Scale-up: bigger CPU	Horizontal scaling (Scale-out: shardin
Transactions	Strict ACID Compliance.	BASE Model (Basically Available, Soft-
Joins	Native SQL `JOIN` operations.	Denormalized data / Application-side j

4. Quick Recall Flow

NoSQL -> Scale-Out Horizontal Architecture -> CAP Theorem Tradeoffs -> Key-Value | Document | Column-Family | Graph